THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **We Distribute, Yet Things Multiply** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** $x^2 + 6x + 9$	$(a+b)^2 = a^2 + 2ab + b^2$
2	**(a)** $x^2 - 4x + 4$	$(a-b)^2 = a^2 - 2ab + b^2$
3	**(a)** $x^2 - 9$	$(a+b)(a-b) = a^2 - b^2$
4	**(a)** $x^2 + 7x + 10$	$(x+a)(x+b) = x^2 + (a+b)x + ab$
5	**(d)** 9999	$(100+1)(100-1) = 100^2 - 1^2$
6	**(c)** 22	Add the two cross products
7	**(b)** $(a-b)^2 = a^2 - 2ab + b^2$	Check by expanding
8	**(b)** $x^2 + 8x + 16$	$(a+b)^2 = a^2 + 2ab + b^2$
9	**(a)** $x^2 - 10x + 25$	$(a-b)^2 = a^2 - 2ab + b^2$
10	**(c)** $x^2 - 36$	$(a+b)(a-b) = a^2 - b^2$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$x^2 + 2 \times x \times 8 + 8^2$ (1) $= x^2 + 16x + 64$ (1).	1+1
12	$6x^2 + 8x + 3x + 4$ (1) $= 6x^2 + 11x + 4$ (1).	1+1
13	$48 \times 52 = (50-2)(50+2)$ (1) $= 2500 - 4 = 2496$ (1).	1+1
14	$x^2 + 2 \times x \times 9 + 9^2$ (1) $= x^2 + 18x + 81$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$(2x)^2 + 2(2x)(3) + 3^2$ (2); $= 4x^2 + 12x + 9$ (1).	2+1
16	$53 = 50 + 3$, $47 = 50 - 3$ (1); product $= 2500 - 9$ (1); $= 2491$ (1).	1+1+1
17	$(x+2)(x+3) = x^2 + 5x + 6$ (1); $(x-2)(x-3) = x^2 - 5x + 6$ (1); sum $= 2x^2 + 12$ (1).	1+1+1
	OR: $(3x)^2 + 2(3x)(4) + 4^2$ (2); $= 9x^2 + 24x + 16$ (1).	2+1
18	$68 = 70 - 2$, $72 = 70 + 2$ (1); product $= 4900 - 4$ (1); $= 4896$ (1).	1+1+1
	OR: $(x+4)(x+5) = x^2 + 9x + 20$ (1); $(x-4)(x-5) = x^2 - 9x + 20$ (1); sum $= 2x^2 + 40$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Area $= (x+3)(x+5)$ (1); $= x^2 + 8x + 15$ (2); at $x = 4$: 63 cm^2 (2).	1+2+2
	OR: $(a+b)^2 = a^2 + 2ab + b^2$, $(a-b)^2 = a^2 - 2ab + b^2$ (2); difference $= 4ab$ (1); for $a = 7$, $b = 3$: $100 - 16 = 84$ and $4ab = 84$ (2).	2+1+2
20	Area $= (x+2)(x+6)$ (1); $= x^2 + 8x + 12$ (2); at $x = 5$: 77 cm^2 (2).	1+2+2
	OR: $(a+b)^2 = a^2 + 2ab + b^2$, $(a-b)^2 = a^2 - 2ab + b^2$ (2); difference $= 4ab$ (1); for $a = 9$, $b = 4$: $169 - 25 = 144$ and $4ab = 144$ (2).	2+1+2

Mistakes to avoid:

- Writing $(a+b)^2 = a^2 + b^2$; the middle term $2ab$ is missed.
- Writing $(a-b)^2 = a^2 - b^2$; correct is $a^2 - 2ab + b^2$, while $a^2 - b^2 = (a+b)(a-b)$.
- Multiplying only the first term of a bracket: $3(x+4) = 3x + 4$ instead of $3x + 12$.
- Sign errors when multiplying: $-2 \times (-3x) = +6x$ and $(x-3)(x-2)$ has $+6$ as the constant term.
- Adding unlike terms: $2x + 3x^2 = 5x^3$; x and x^2 are not like terms.
- Squaring a product without squaring the coefficient: $(3x)^2 = 9x^2$, not $3x^2$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **We Distribute, Yet Things Multiply** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** $x^2 + 10x + 21$	$(x + a)(x + b) = x^2 + (a + b)x + ab$
2	**(d)** 9996	$(100 + 2)(100 - 2) = 100^2 - 2^2$
3	**(b)** 20	Add the two cross products
4	**(c)** $(a + b)(a - b)$	Difference of squares
5	**(d)** $x^2 + 10x + 25$	$(a + b)^2 = a^2 + 2ab + b^2$
6	**(a)** $x^2 - 16x + 64$	$(a - b)^2 = a^2 - 2ab + b^2$
7	**(d)** $x^2 - 49$	$(a + b)(a - b) = a^2 - b^2$
8	**(d)** $x^2 + 10x + 24$	$(x + a)(x + b) = x^2 + (a + b)x + ab$
9	**(a)** 9991	$(100 + 3)(100 - 3) = 100^2 - 3^2$
10	**(a)** 22	Add the two cross products

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$3x^2 + 15x + 2x + 10$ (1) $= 3x^2 + 17x + 10$ (1).	1+1
12	$97 \times 103 = (100 - 3)(100 + 3)$ (1) $= 10000 - 9 = 9991$ (1).	1+1
13	$x^2 + 2 \times x \times 11 + 11^2$ (1) $= x^2 + 22x + 121$ (1).	1+1
14	$8x^2 + 4x + 6x + 3$ (1) $= 8x^2 + 10x + 3$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$(5x)^2 + 2(5x)(1) + 1^2$ (2); $= 25x^2 + 10x + 1$ (1).	2+1
16	$91 = 100 - 9$, $109 = 100 + 9$ (1); product $= 10000 - 81$ (1); $= 9919$ (1).	1+1+1
17	$(x + 1)(x + 7) = x^2 + 8x + 7$ (1); $(x - 1)(x - 7) = x^2 - 8x + 7$ (1); sum $= 2x^2 + 14$ (1).	1+1+1
	OR: $(4x)^2 + 2(4x)(5) + 5^2$ (2); $= 16x^2 + 40x + 25$ (1).	2+1
18	$35 = 40 - 5$, $45 = 40 + 5$ (1); product $= 1600 - 25$ (1); $= 1575$ (1).	1+1+1
	OR: $(x + 3)(x + 6) = x^2 + 9x + 18$ (1); $(x - 3)(x - 6) = x^2 - 9x + 18$ (1); sum $= 2x^2 + 36$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Area $= (x + 4)(x + 7)$ (1); $= x^2 + 11x + 28$ (2); at $x = 3$: 70 cm^2 (2).	1+2+2
	OR: $(a + b)^2 = a^2 + 2ab + b^2$, $(a - b)^2 = a^2 - 2ab + b^2$ (2); difference $= 4ab$ (1); for $a = 8$, $b = 5$: $169 - 9 = 160$ and $4ab = 160$ (2).	2+1+2
20	Area $= (x + 1)(x + 9)$ (1); $= x^2 + 10x + 9$ (2); at $x = 6$: 105 cm^2 (2).	1+2+2
	OR: $(a + b)^2 = a^2 + 2ab + b^2$, $(a - b)^2 = a^2 - 2ab + b^2$ (2); difference $= 4ab$ (1); for $a = 6$, $b = 2$: $64 - 16 = 48$ and $4ab = 48$ (2).	2+1+2

Mistakes to avoid:

- Writing $(a + b)^2 = a^2 + b^2$; the middle term $2ab$ is missed.
- Writing $(a - b)^2 = a^2 - b^2$; correct is $a^2 - 2ab + b^2$, while $a^2 - b^2 = (a + b)(a - b)$.
- Multiplying only the first term of a bracket: $3(x + 4) = 3x + 4$ instead of $3x + 12$.
- Sign errors when multiplying: $-2 \times (-3x) = +6x$ and $(x - 3)(x - 2)$ has $+6$ as the constant term.
- Adding unlike terms: $2x + 3x^2 = 5x^3$; x and x^2 are not like terms.
- Squaring a product without squaring the coefficient: $(3x)^2 = 9x^2$, not $3x^2$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **We Distribute, Yet Things Multiply** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** $x^2 + 8x + 16$	$(x+4)^2$
2	**(d)** $x^2 + 12x + 36$	$(a+b)^2 = a^2 + 2ab + b^2$
3	**(d)** $x^2 - 18x + 81$	$(a-b)^2 = a^2 - 2ab + b^2$
4	**(b)** $x^2 - 121$	$(a+b)(a-b) = a^2 - b^2$
5	**(d)** $x^2 + 9x + 8$	$(x+a)(x+b) = x^2 + (a+b)x + ab$
6	**(a)** 9984	$(100+4)(100-4) = 100^2 - 4^2$
7	**(b)** 25	Add the two cross products
8	**(b)** the difference of the squares of a and b	Standard identity
9	**(c)** $x^2 + 14x + 49$	$(a+b)^2 = a^2 + 2ab + b^2$
10	**(b)** $x^2 - 20x + 100$	$(a-b)^2 = a^2 - 2ab + b^2$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$45 \times 55 = (50-5)(50+5)$ (1) $= 2500 - 25 = 2475$ (1).	1+1
12	$x^2 + 2 \times x \times 12 + 12^2$ (1) $= x^2 + 24x + 144$ (1).	1+1
13	$15x^2 + 10x + 6x + 4$ (1) $= 15x^2 + 16x + 4$ (1).	1+1
14	$86 \times 94 = (90-4)(90+4)$ (1) $= 8100 - 16 = 8084$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$(2x)^2 + 2(2x)(7) + 7^2$ (2); $= 4x^2 + 28x + 49$ (1).	2+1
16	$74 = 80 - 6$, $86 = 80 + 6$ (1); product $= 6400 - 36$ (1); $= 6364$ (1).	1+1+1
17	$(x+2)(x+8) = x^2 + 10x + 16$ (1); $(x-2)(x-8) = x^2 - 10x + 16$ (1); sum $= 2x^2 + 32$ (1).	1+1+1
	OR: $(3x)^2 + 2(3x)(2) + 2^2$ (2); $= 9x^2 + 12x + 4$ (1).	2+1
18	$118 = 120 - 2$, $122 = 120 + 2$ (1); product $= 14400 - 4$ (1); $= 14396$ (1).	1+1+1
	OR: $(x+5)(x+4) = x^2 + 9x + 20$ (1); $(x-5)(x-4) = x^2 - 9x + 20$ (1); sum $= 2x^2 + 40$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Area $= (x+5)(x+8)$ (1); $= x^2 + 13x + 40$ (2); at $x = 2$: 70 cm^2 (2).	1+2+2
	OR: $(a+b)^2 = a^2 + 2ab + b^2$, $(a-b)^2 = a^2 - 2ab + b^2$ (2); difference $= 4ab$ (1); for $a = 10$, $b = 3$: $169 - 49 = 120$ and $4ab = 120$ (2).	2+1+2
20	Area $= (x+3)(x+4)$ (1); $= x^2 + 7x + 12$ (2); at $x = 10$: 182 cm^2 (2).	1+2+2
	OR: $(a+b)^2 = a^2 + 2ab + b^2$, $(a-b)^2 = a^2 - 2ab + b^2$ (2); difference $= 4ab$ (1); for $a = 12$, $b = 5$: $289 - 49 = 240$ and $4ab = 240$ (2).	2+1+2

Mistakes to avoid:

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- Writing $(a-b)^2 = a^2 - b^2$; correct is $a^2 - 2ab + b^2$, while $a^2 - b^2 = (a+b)(a-b)$.
- Multiplying only the first term of a bracket: $3(x+4) = 3x + 4$ instead of $3x + 12$.
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